Sergei Volkov

List of Publications & Research Contributions

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ORCID: 0000-0002-5385-1541 · GitHub: github.com/taranarmo
Domain Expertise: Limnology, Hydrophysics, Convective Turbulence, Environmental Data Systems

PEER-REVIEWED JOURNAL PUBLICATIONS

2021 **Full Reynolds Stress Tensor of Convective Turbulence Estimated with Paired Acoustic Doppler Current Profilers**
S. Bogdanov, G. Kirillin, S. Volkov, G. Zdorovennova
Earth and Space Science Open Archive (ESSOAr) · DOI: 10.1002/essoar.10507975.1

2019 **Fine scale structure of convective mixed layer in ice-covered lake**
S. Volkov, S. Bogdanov, R. Zdorovennov, G. Zdorovennova, A. Terzhevik, N. Palshin, D. Bouffard, G. Kirillin
Environmental Fluid Mechanics, 19(5), 1151–1165 · DOI: 10.1007/s10652-018-9652-2

2019 **Under-ice convection dynamics in a boreal lake**
D. Bouffard, G. Zdorovennova, S. Bogdanov, T. Efremova, S. Lavanchy, N. Palshin, A. Terzhevik, L. Vinnå, S. Volkov, A. Wüest, R. Zdorovennov, H. Ulloa
Inland Waters, 9(2), 142–161 · DOI: 10.1080/20442041.2018.1533356

2019 **Structure and dynamics of convective mixing in Lake Onego under ice-covered conditions**
S. Bogdanov, G. Zdorovennova, S. Volkov, R. Zdorovennov, N. Palshin, T. Efremova, A. Terzhevik, D. Bouffard
Inland Waters, 9(2), 162–178 · DOI: 10.1080/20442041.2018.1551655

2018 **Albedo of a Small Ice-Covered Boreal Lake: Daily, Meso-Scale and Interannual Variability on the Background of Regional Climate**
G. Zdorovennova, N. Palshin, T. Efremova, R. Zdorovennov, G. Gavrilenko, S. Volkov, S. Bogdanov, A. Terzhevik
Geosciences, 8(6), 206 · DOI: 10.3390/geosciences8060206

INTERDISCIPLINARY RESEARCH PROJECTS

Lake Pycnoclines Trap Organic Particles Forming Hot Spots of Accelerated Carbon Cycling (PycnoTrap) 2021 – 2023
Leibniz Institute of Freshwater Ecology and Inland Fisheries (IGB Berlin)

Joint limnological project researching sinking velocity, biomass degradation, and hydrodynamic particle trapping across pycnoclines in stratified lake columns. Designed twin-action underwater camera frames and built computer vision / stereo-video tracking pipelines to calculate 3D velocity vectors and estimate vertical particulate carbon flux.

Lake Onego: Life Under the Ice 2015 – 2017
Northern Water Problems Institute (NWPI) / EPFL Limnology Center · Project Website

International multi-institutional field campaigns investigating under-ice convective mixing, thermal boundary regimes, and winter ecosystem dynamics in Europe's second-largest boreal lake.

ACADEMIC THESES & TEACHING

Specialist Diploma Thesis (M.Sc. Equivalent): *"Noniterative Heat Exchanger Calculation"* 2015
Supervisor: Prof. Sergei Bogdanov, Petrozavodsk State University
Proposed a direct analytical calculation methodology for heat exchanger design, avoiding non-convergent numerical optimization algorithms.

Teaching Assistant: Department of Energy Supplies, Petrozavodsk State University 2016 – 2020
Delivered lectures and laboratory seminar instruction in Thermodynamics, Heat and Mass Transfer, and Applied Fluid Mechanics.